Experiment - II

AIM :- To convert an ER diagram into a relational schema and implement it using MySQL

Relational Schema :-

Customer (customer_ID, Name, Email, Phone)

Product (Product_ID, Category_Name, Price, Category_ID)

Category (category_ID, category_Name)

Orders (order_ID, order_Date, Customer_ID)

Order_Item (order_ID, Product_ID, Quantity Price)

Source Code :

```
CREATE TABLE CUSTOMER (CUSTOMER_ID, INT PRIMARY KEY
NAME VARCHAR(50) NOTNULL,
E-MAIL VARCHAR (100) UNIQUE;)

CREATE TABLE CATEGORY(
CATEGORY_ID INTPRIMARY KEY,
CATEGORY_NAME VARCHAR(50));

CREATE TABLE PRODUCT(
PRODUCT_ID INT PRIMARY KEY,
PRODUCT_NAME VARCHAR (50),
PRICE DECIMAL (10.1);
CATEGORY_ID INT,
FOREIGN KEY (category_ID)
    REFRENCES CATEGORY (category_ID));
```

```
CREATE TABLE ORDERS(
   ORDER_ID INT PRIMARY KEY,
   ORDER_DATE DATE
   CUSTOMER_ID INT,
   FOREIGN KEY(CUSTOMER_ID)
   REFERENCES CUSTOMER(CUSTOMER_ID));
CREATE TABLE ORDER_ITEM(
   ORDER_ID INT,
   PRODUCT_ID INT,
    QUANTITY INT,
   PRICE DECIMAL(10.2),
  PRIMARY KEY((ORDER_ID), PRODUCT_ID);
   FOREIGN KEY(ORDER_ID)
   REFERENCES ORDERS(ORDER_ID)

CREATE TABLE ADDRESS(
    ADDRESS_ID INT PRIMARY KEY,
    ADDRESS_type VARCHAR(20) NOT NULL,
    HOUSE-NO VARCHAR(20) NOT NULL,
    STREET VARCHAR(100) NOT NULL,
    CITY VARCHAR(100) NOT NULL,
    STATE VARCHAR(50) NOT NULL,
    PINCODE VARCHAR(10) NOT NULL);
CREATE TABLE PAYMENT(
   PAYMENT_ID INT PRIMARY KEY, ORDER_ID INT NOT NULL,
   PAYMENT_Method VARCHAR(30) NOT NULL, Amount DECIMAL(12,2) NOT NU
   PAYMENT_Status VARCHAR(30) NOT NULL,
   TRANSACTION_ID VARCHAR(100) UNIQUE,
   FOREIGN KEY(Order_ID) REFERENCES ORDER(order_ID));
```

```
CREATE TABLE DELIVERY(
  DELIVERY_ID INT PRIMARY KEY,
  ORDER_ID INT NOT NULL,
  ADDRESS_ID INT NOT NULL,
  DELIVERY_STATUS VARCHAR(30) NOT NULL,
  TRACKING_ID  VARCHAR(100) UNIQUE,
  FOREIGN KEY (order_ID) REFERENCE order (ORDER_ID));

CREATE TABLE ELECTRONICS(
  PRODUCT_ID INT PRIMARY KEY,
  BRAND VARCHAR(100) NOT NULL,
  WARRENTY_PERIOD VARCHAR(50),
  FOREIGN KEY (PRODUCT_ID) REFERENCES Product(PRODUCT_ID));

CREATE TABLE FASHION(
  PRODUCT_ID INT PRIMARY KEY,
    SIZE VARCHAR(20) NOT NULL,
    COLOUR VARCHAR(30) NOT NULL,
  FOREIGN KEY (PRODUCT_ID) REFERENCES PRODUCT(PRODUCT_ID));

INSERT INTO CUSTOMER VALUES
  (1,'Rahul Sharma','rahul@gmail.com','rahul123','2026-01-10','2005-05-16',
   '12A','MGROAD','Delhi','Delhi','110001');

INSERT INTO CUSTOMER_PHONE VALUES
   (1,'9876543210'), (1,'9123456780');

INSERT INTO SELLER VALUES
  (101,'Tech World India','Tech@gmail.com', 9876500000);

INSERT INTO SELLER_BANK_ACCOUNT VALUES
  (101,'BANK1001');

INSERT INTO CATEGORY VALUES
   (1,'Electronics','Electronic products');
   (2,'Fashion','Fashion products');
```

```
INSERT INTO PRODUCT VALUES
  (1001, 'Laptop', 'Gaming Laptop', 65000.00, 10, 101, 1);
  (1002, 'T-shirt', 'Cotton-shirt', 799.00, 50, 101, 2);
INSERT INTO ELECTRONICS VALUES
  ('1001', 'Dell', '2 Years');
INSERT INTO Fashion VALUES
  (1002, 'M', 'Black');
INSERT INTO ORDER VALUES
  (5001, 1, '2026-08-19', 'Confirmed', 65799.00);
INSERT INTO ORDER_Item VALUES
  (5001, 1001, 1, 65000.00, 1000.00),
  (5001, 1002, 1, 799.00, 0.00);
INSERT INTO ADDRESS VALUES
  (201, 'Home', '12A', 'MGROAD', 'DELHI', 'Delhi', '110001');
INSERT INTO PAYMENT VALUES
  (3001, 5001, 'UPI', 65799.00, 'Success', 'TXN001');
INSERT INTO Delivery VALUES
  (4001, 5001, 201, 'In Transit', TRK001);

SELECT * FROM CUSTOMER;
SELECT * FROM SELLER;
SELECT * FROM PRODUCT;
SELECT * FROM ORDER;
SELECT * FROM ORDERITEM;
SELECT * FROM PAYMENT;
SELECT * FROM DELIVERY;
```

